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CHAPTER 14

Toward Automating Causal Discovery in Financial Markets and Beyond

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This paper introduces a novel machine learning (ML) framework for causal discovery based on recent advances in large language models (LLMs) and discusses the applications of these causal discovery techniques to investment management. Unlike typical data-driven methods for data discovery, the framework using the implicit "world knowledge" in state-of-the-art LLMs to automate the expert judgment approach to causal discovery. A key application that is explored in detail is end-to-end causal factor analysis, where the authors demonstrate the utility of our method in specifying and analyzing detailed causal models for financial markets. This paper also conducts a comparative analysis, juxtaposing the new approach with conventional methods, to underscore the enhanced capability of the framework in revealing intricate causal dynamics in financial data.

Key Takeaways:

(1) The authors propose and implement an end-to-end method for causal modeling with applications to financial markets, from causal discovery to causal scenario modeling.
(2) The authors demonstrate the application of the end-to-end framework to factor investing and how the framework can be applied to different sets of input features.
(3) The authors show that large language models have taken a significant step forward toward automating causal discovery in finance and potentially across many more diverse business domains.

14.1. Introduction

In recent years, the intersection of machine learning (ML) and causal inference has emerged as a vital research area, promising to unveil deeper insights into the causal structures within complex datasets. Traditional approaches to causal discovery often struggle with the intricacies and nonlinear relationships present in real-world data. In addition, ML techniques have historically lacked a "world model," which meant an over-reliance on statistical modeling and observed effects and difficulty in overlaying priors regarding causal relationships in an automated fashion. To address these challenges, the authors introduce a novel ML framework for causal discovery, designed to be versatile and applicable across a spectrum of domains, by leveraging recent advances in large language models (LLMs).

Factor investing, which involves identifying and leveraging specific factors or drivers behind asset returns, serves as a robust testbed to showcase the efficacy of the framework. The finance industry, characterized by its complex and dynamic data, demands robust methods for causal analysis to enhance investment strategies and risk management. Unlike conventional correlation-based analyses, this paper looks at creating alternate systematic approaches for selecting factors for constructing factor models.

The paper outlines the process of employing this ML-driven framework for causal discovery in factor investing. The authors begin with the formulation of causal models and applications of LLMs to their formulation (causal discovery). This is followed by an end-to-end experiment for how an LLM-assisted causal discovery approach can aid in creating causal factor models, as well as a demonstration of how the framework can be easily applied to different feature sets.

A key aspect of the end-to-end application is the analysis of factor betas, their stability, and their significance in the causal framework. The authors utilize do-calculus for conducting simulations that assess the causal impact of various factors on asset returns. As with traditional causal investing frameworks, these simulations provide insights into the potential effects of market shifts or strategic interventions.

The structure of the paper is as follows: Section 14.2 provides the theoretical background and reviews related work, Section 14.3 details the methodology, and Section 14.5 presents the application in factor investing with results. Section 14.6 concludes the paper and provides potential directions of future work.

14.2. Literature Review

14.2.1. *Causal modeling in finance*

The field of causal modeling is becoming of greater importance in finance and investment management, addressing the need for formulating robust economic theories to counteract the ease with which financial strategies that perform well on a backtest but may not generalize well into the future can be found. The advent of ML and big data has accelerated the trend of false investment theories being identified in research as shown in de Prado [1], and causal methods in finance have been proposed as a general methodology for making the search of investment strategies more robust as seen in Prado [2].

In classical approaches to factor investing, such as Fama and French [3] and Fama and French [4], the authors theorize that simple models of several *factors* can explain components of equity returns. These methods are predicated on hidden hypothesis of causal relationships between the factors used and returns. On the other hand, the betas that are computed in practice using supervised techniques such as ordinary least squares (OLS) are based on correlation and represent an associational rather than causal relationship, and so most of the time it is an unstable estimated value that can result in "spurious" findings in econometrics and finance. As in Prado [2], causal frameworks for factor investing have been proposed in which an explicit causal relationship between factors and returns of assets, formulated as a *directed acyclic graph (DAG)*, is the starting point. These frameworks make the pre-assumed hypotheses of causality falsifiable by causal discovery algorithms or experts' opinions.

Classical approaches to this problem include data-driven methodologies as seen in Zhang et al. [5], where the authors claim that classical feature selection algorithms in quantitative finance like stepwise regression analysis (SRA), principal component analysis (PCA), decision tree (DT), and information gain are based on correlation, and so they cannot represent the causal direct influence of features on assets' returns. They propose the causal feature selection (CFS) algorithm, and they show that their CFS method improves feature selection in both accuracy-based and investment metrics.

14.2.2. *Causal discovery in finance*

Causal discovery is a core aspect of causal modeling approaches, and causal discovery in finance holds a particularly important role due to the lack of consensus around causal relationships in economics and the widespread use of novel datasets and factors.

In Polakow et al. [6], the authors take a philosophical view toward the rising number of applications of causal inference in quantitative finance. They argue that these applications may be limited due to the complexity and reflexivity of the financial markets. First, causal graphs can be verified ex-post and models being used ex-ante for prediction need calibration. Second, as financial markets are not a closed system, the causal mechanism in the market may change over time. Also, due to the reflexivity of the market and different models that financial activists use, the arrow of causation may change in different periods. This increases the need for more automated and dynamic approaches for causal discovery and combinations of "first-principles" and data-driven causal discovery methods; the framework shows an end-to-end automation for both.

In Hao et al. [7], the authors explain the limitations of data-based causal discovery algorithms in high-dimension data sets, which is the case in finance, due to the reduction of accuracy and increase of computation complexity. They mention two main approaches of causal discovery algorithms: Structure learning approaches and direction learning approaches. They classify the structure learning approaches into two main groups: *constraint-based* *methods* like PC and FCI and *search & score methods* like Bayesian information criterion (BIC). The problem with these methods is that they only identify the DAG up to the Markov equivalence class. The conditional independence (IC) test that these methods are based on, in high-dimensional data, becomes extremely difficult, and searching space of the DAG becomes super-exponential in the number of variables. For the directional learning approaches, that are based on additive noise models (ANM), they mention two *LINGAM* and *Information-Geometric Causal Inference* (IGCI) algorithms that decrease the time complexity of causal discovery but work only on low-dimensional data sets. To overcome the curse of dimension in the causal discovery method, they introduce a three-step algorithm, the *causal discovery in high dimension* (CDHL).

In a more recent attempt for causal discovery in a high-dimensional data set, in Hasan and Gani [8], the authors remind the necessity of causal structure in causal ML for simulating the data generation process. They also mention the difficulty of the causal discovery task for high-dimensional data. As a solution, they propose imposing a prior on the structure of the DAG from prior knowledge, expert opinion, or other information sources. To use this prior knowledge for causal discovery from data, they introduce a reinforcement algorithm that penalizes the causal discovery algorithm for deviating from priors. Thus, they offer a systematic way to impose priors on the DAG.

14.2.3. *Large language models for causal discovery*

LLMs have recently begun seeing widespread use for causal discovery. In Lampinen et al. [9], the authors deal with the question of whether a passive learner can learn causal intervention strategies that it can use in the test data to uncover its underlying causal structure. They designed a two-stage agent that in the first stage (*experimentation strategy*), intervenes in the data generation process and learns the causal structure, and in the second stage (*exploitation strategy*), it uses its causal knowledge to intervene in the remaining of the episode to achieve a goal g. To learn the experimentation part, they use a large language model (a 70 billion parameter Chinchilla) and train it by the next word prediction task. They deduce that despite confounding variables in the training data set and passive learning, the passive learner agent can come up with causal discovery tasks in the test data, though an active RL learner with human feedback on on-policy data may improve the results.

In Zhang et al. [10], authors mention three types of causal inference that one may expect from a LLM. Type I is to identify the causal relationship between two subjects using domain knowledge. Type II is causal inference from a data set, and for example, deciding which variable is the cause of the other. Type III is the quantitative estimation of the effect of a cause variable on a dependent variable. This paper shows several practical improvements for prompt engineering strategies to determine Type I causal relationships, as well as combining LLMs for causal discovery with existing methodoloigies to create an end-to-end system that can combine all three forms of causal inference.

In Jin et al. [11], authors construct a verbalized task of inferring causation from correlations, CORR2CAUS, using conventional causal discovery methods like Peter–Clark (PC), which consists of 400K data points. By prompting this task to 17 main LLMs, they deduce that the outputs are almost random for all language models, and none of them are able to infer causality from correlation structure. By fine-tuning these models with this task, the results improve slightly but still not significantly. This paper shows that the usage of current state-of-the-art LLMs (GPT-4) combined with robust prompt engineering strategies can generate causal inferences that are of practical use.

In Kıcıman et al. [12], the authors take a positive attitude toward causal discovery and causal reasoning by LLMs. They divide causal discovery from two points of view. The first angle is *covariance vs. logic-based causality*. Covariance causality is the causal discovery using statistical methods and numerical evidence to discover the existence and strength of a causal relationship between two variables. While in a logic-based approach, it uses logical reasoning and domain expertise to discover a causal relationship. The other view is *type vs. actual causality*. In type causality, it tries to perform a causality inference such as causal discovery or causal effect estimation between two variables. While actual causality probes for causal reasons for a special event. The authors also consider various causal tasks, including *causal discovery*, *effect inference* and *attribution*. Using GPT-3.5 and GPT-4, the authors claim that they could achieve a higher accuracy in various causality tasks. They've achieved 97% accuracy for pairwise causal discovery, 92% accuracy in logic-based (or counterfactual) reasoning tasks, and 86% accuracy in actual causality tasks, which improve the current causal discovery algorithms by more than 10%. Despite these improvements, LLMs make some trivial mistakes that open new doors for further research and developments in the field of causal reasoning by LLMs.

In Naik et al. [13], the authors use LLMs to draw causal edges between 18 medical features related to lung cancer. The authors mention the lack of use of domain experts in usual score-based or constraint-based causal discovery methods that can be filled by using LLMs for achieving DAGs. They compare the DAGs generated by LLMs in both edge-by-edge prompting or prompting for the whole graph at once, and they achieve higher accuracy with respect to usual methods of causal discovery when compared based on the Bdeu score. They also mention that the accuracy might even increase if the LLM was trained over a specialized database. The framework presented within this paper allows for DAG generation that is more robust and scalable to hundreds of input features, as detailed in Section 14.3.

In Long et al. [14], the authors mention that usual causal discovery methods have the limitation that they only find the DAG up to *Markov equivalence class (MEC)*. And then an expert can choose the right DAG from the MEC. For cases when experts can make mistakes, they formalize the question of finding the true DAG among MEC by an *imperfect expert* as minimizing the size of the MEC chosen by the usual causal discovery method while the probability of inclusion of the true DAG is controlled and superior to a constant. Then they use an LLM as an imperfect expert that mitigates the performance. They suggest using a Bayesian causal discovery approach and replacing the MEC-based prior with a learned posterior distribution over graphs. This paper uses similar ideas by splitting the DAG generation task into sub-graphs, as detailed in Section 14.3.2.

While the articles just mentioned use LLMs as domain experts to deduce causal relationships between variables, in Ban et al. [15], the authors combine the classical score-based causal structural learning (CSL) methods with LLMs for obtaining the DAG for a data set. They query various LLMs, including GPT-4, for causal relationships between pairs of variables, giving a short description of the variables. Then they use the LLMs response as a prior for score-based CSL algorithms in two hard and soft settings. In the hard setting, they enforce the causal ancestry relations proposed by the LLM to be completely satisfied by the set of Bayesian networks they search in. In the soft setting, they consider a penalty for deviating from the prior conditions but let some of the LLMs suggestions be violated. The authors report that they have achieved a higher accuracy for the causal discovery task and drawing the true DAG, when using the LLMs as a prior for CSL algorithms compared to the case that they run the CSL algorithms directly on the data without LLMs suggestions as a prior.

In Hollmann et al. [16], the authors propose using LLMs for *Context-Aware Automated Feature Engineering (CAAFE)*. They mention that the most time-consuming tasks of a data scientist are data cleaning and feature engineering, and most data engineering algorithms come up with disintuitive features. So they propose CAAFE as a pipeline for automatic feature engineering empowered by vast knowledge of LLMs. They describe the task and the data set by a prompt to an LLM. The LLM generates a Python code that transforms the initial raw data into processed features that an AutoML system can use as input data. It also describes the features that are extracted in terms of natural language that increases the interpretability of the AutoML pipeline. Although they increase the ROC AUC by this method, LLMs' hallucinations are still pitfalls of this automated feature engineering method. This paper utilizes some similar ideas for constructing different features to represent graph edges in the causal DAGs, as detailed in Section 14.4.2.

Overall, there has been a significant uptick in research for LLMs for causal discovery and modeling, but this paper provides perhaps the most complete overview to date of an end-to-end causal discovery and modeling framework applicable in the finance and investment management domains.

14.2.4. *Large language models for quantitative finance*

The field of finance has been influenced by the introduction of *the natural language processing* (NLP) method by sentiment analysis and nowcasting. With the introduction of LLMs, finance has been considered one of the first fields to benefit from, and the rising number of articles related to the applications of LLMs in quantitative finance in recent years just shows this. However, using LLMs that are trained on general databases and even Fin-LLMs that are trained on financial data in financial tasks may lead to challenges reviewed in this section, including availability, output volatility, and hallucinations.

In Liu et al. [17], the authors propose the problems of using LLMs trained on general text data for applications in finance. The difficulty might be in the different meanings or sentiments of words in general and financial contexts or the dilution of financial data in a general database. There have been some efforts, like BloombergGPT to train LLM models on financial data that have overcome general LLMs in financial tasks. But lack of access to their database and API's to the public makes it difficult to use them for research and development purposes and making contributions. In addition, training their model is expensive, and for real applications in finance, we'd need to fine-tune the model on new and live data. So the authors propose *Financial generative pre-trained transformer* (Fin-GPT) as a solution for democratizing financial LLMs. They provide financial data from 34 different data sources regularly updated through API. They also introduce reinforcement learning with stock prices as a method for fine-tuning LLMs with market data. They show empirically the effectiveness of their method in various financial tasks.

In Yu [18], the author studies the effect of uncertainty in the LLMs and its impact on decision-making in finance. The uncertainty in the output of an LLM might be due to the temperature setting or the ambiguity of or change in the query. For example, in the task of sentiment classification, the sentence might not have a clear positive or negative sentiment. In this article, the author proposes a way to quantify the uncertainty and volatility of an LLM in an investment strategy without direct access to intermediate layers of the LLM in an LLM-based trading strategy. This quantification can be used to enhance the uncertainty effects in the financial decision-making or trading strategy.

In Kang and Liu [19], the authors study the hallucination as a deficiency in financial LLMs (FinLLMs). Hallucination is a main issue in financial tasks due to the intricate nature of financial concepts. They empirically measure the performance of FinLLMs in learning and memorizing financial concepts. They also measure the performance of LLMs for a basic financial task, say querying price data. They also investigate four methods to enhance LLMs' hallucination in finance, say *Few Shot Prompting*, *Decoding by Contrasting Layers* (DoLa), *the Retrieval Augmentation Generation method* (RAG), and *the prompt-based tool learning method*. However, they remind the necessity for more research on LLMs' hallucination issues in finance.

14.3. Methodology

The initial phase of the causal discovery framework involves the construction of a DAG, $\mathscr{G} = (\mathscr{V}, \mathscr{E})$, to represent the presumed causal relationships among the variables. In this context, each node in $\mathscr{G}$ symbolizes a distinct variable, such as stock returns, economic indicators, or company fundamentals. The directed edges in $\mathscr{E}$ indicate the hypothesized causal directions. For instance, if variable X (e.g., a company's earnings) is conjectured to causally influence variable Y (e.g., stock returns), this relationship is depicted with a directed edge from X to Y in the DAG.

Upon establishing the DAG, the subsequent step entails utilizing observational data to deduce the strengths or weights of the defined causal relationships. This process typically employs statistical methodologies like regression analysis. For a simple scenario where Y is causally impacted by X, the regression model can be formalized as:

$$Y = \beta_0 + \beta_1 X + \epsilon,$$

where β_0 represents the intercept, β_1 is the coefficient indicating the causal effect of X on Y, and ϵ denotes the error term. The primary objective is to accurately estimate β_1, thereby quantifying the causal impact of X on Y. In essence, the DAG functions as a pre-modeling feature selection mechanism, guiding the formulation of the regression model.

The concept of interventions, operationalized through the *do operation*, is pivotal in simulating the influence of external manipulations on the variables. For example, to explore the effect on stock returns (Y) following a hypothetical increase in a company's earnings (X), the *do operation*, denoted as $\mathrm{do}(X = x)$, is applied. This operation effectively severs any incoming causal links to X in the DAG, rendering X independent of its usual precursors. The distribution of Y under this intervention is denoted as $P(Y \mid \mathrm{do}(X = x))$, representing the distribution of Y when X is externally set to a particular value x.

In the context of a simple causal relationship where X directly influences Y without confounders, $P(Y \mid \mathrm{do}(X = x))$ can be directly inferred from the regression model by substituting x for X. In more intricate scenarios involving confounders or indirect causal paths, additional adjustments and analytical steps are necessitated to accurately compute $P(Y \mid \mathrm{do}(X = x))$ using do-calculus rules, assuming a correctly specified causal model.

14.3.1. *LLMs for causal discovery*

The authors propose the following general framework for automated causal DAG construction based on any arbitrary set of features for a supervised learning task with the single constraint being each feature must have a valid name and definition.

Though recent advances in LLMs have greatly increased the length of the input context window, the longest context-length models are in practice not always the most performant for highly complex tasks. Due to the complexity of causal discovery and the nondeterministic behavior of LLMs, the authors have observed the following current limiting factors in causal DAG generation:

(1) Current state of the art LLMs have limited bandwidth relating to the quantity of features that can be evaluated simultaneously. Indeed, the authors have observed a drop in causal inference quality when the input context contained upward of 30 features for experiments with GPT-4.
(2) The set of causal relationships generated for each LLM inference is not unique.

In order to tackle these limitations, the authors split the causal discovery method into two distinct phases: Feature clustering and causal discovery. The overall process is described below:

14.3.2. *LLMs for feature clustering*

The initial step in the proposed causal discovery process is splitting the input features F into distinct groups of at most n_{feat} features and generating a causal DAG for each group. This is a solution to the drop off in causal inference quality seen with inputting large numbers of features into a single forward pass for an LLM call. For the experiments in this paper, the authors set $n_{feat} = 30$ as that was the threshold after which the quality and consistency of causal relationships generated was observed to drop off significantly.

The authors propose as an alternative to classical correlation clustering algorithms using LLMs to cluster the input features into mutually exclusive groups. This process proceeds in three steps, implemented as three distinct styles of prompt. This approach helps greatly scale the number of input features F that a DAG can be constructed for. This approach can be generalized to work in a loop (applying Prompt 1 iteratively until the subgroups are small enough to apply Prompt 2) to scale up the number of input features even further.

Prompt 1—pre-clustering. The input features F are grouped into an initial grouping denoted as G_1. Each group g in G_1 contains a subset of features in F, such that $g \subset F$ and $G_1 = \{g_1, g_2, ..., g_n\}$, where n is the number of initial groups.

Prompt 2—creates sub-groups: For each group g in G_1, a set of subgroups is created. Denote the set of subgroups created from group g as S_g, where $S_g = \{s_{g1}, s_{g2}, ..., s_{gm}\}$ and m is the number of subgroups for group g. Each subgroup s is a subset of the parent group g, such that $s \subset g$.

Prompt 3—final clustering: Each subgroup s in each S_g is then used as the element in another clustering process. The result of this clustering process is the final grouping, denoted as G_f. Each group in G_f is formed by clustering the subgroups S_g from the initial groups G_1.

The final clustering is exhaustive and mutually exclusive. Now each group in G_f has an LLM-generated name and description. Thus, if necessary, one can combine groups together very easily to obtain a desired number of clusters. One can always also recluster groups in G_f if he wants more granular clusters. This technique thus provides a lot of liberty to the user.

The authors first construct a prompt that utilizes the names and definitions of each feature in F to separate them into n distinct groups or clusters $\{g_1, ..., g_n\}$.

Using the names and definitions of each feature, they construct a prompt to return another clustering of the initial groups $G_1 = \{g_1, ..., g_n\}$ into distinct sub-groups $S_g = \{s_{g1}, s_{g2}, ..., s_{gm}\} \forall g \in G_1$ of at most n_{feat} features each based on the definitions provided for each feature.

The authors compare the results obtained using a classical clustering algorithm versus the proposed method in the Applications section. Depending on the number of features and the average length of the feature names and definitions, it may be needed initially to partition the set of features into random distinct groups in order to fit the context length of the LLM. If the initial separation is needed, the intermediate groups S_g regrouped them all using the final prompt (Prompt 3). The authors also include a sample Prompt 4, which could be used to combine clusters using their names and definitions if G_f is considered too granular.

14.4. DAG Generation—LLM for Causal Discovery

Having generated the final set of groupings G_f, the authors now focus on the generation of the causal DAG modeling the intra-causal and inter-causal relationships between these groups.

In order to generate a valid final DAG, using the names and descriptions of the set of features, all possible causal relationships in each group of features are iteratively generated and modeled as DAGs. Subsequent prompts are then used to validate and refine each of the sub-graph DAGs obtained in this step, similar to the chain-of-thought technique seen in Wei et al. [20].

To model the inter-group causal relationships, the authors start by generating descriptions for each group based on the definitions of the features belonging to each cluster in G_f. Using these descriptions, they generate the inter-group causal DAGs. Finally, the causal relationships present in this inter-group DAG are used to link together each of the features belonging to each group. A visualization explaining this last step can be seen in Fig. 14.1.

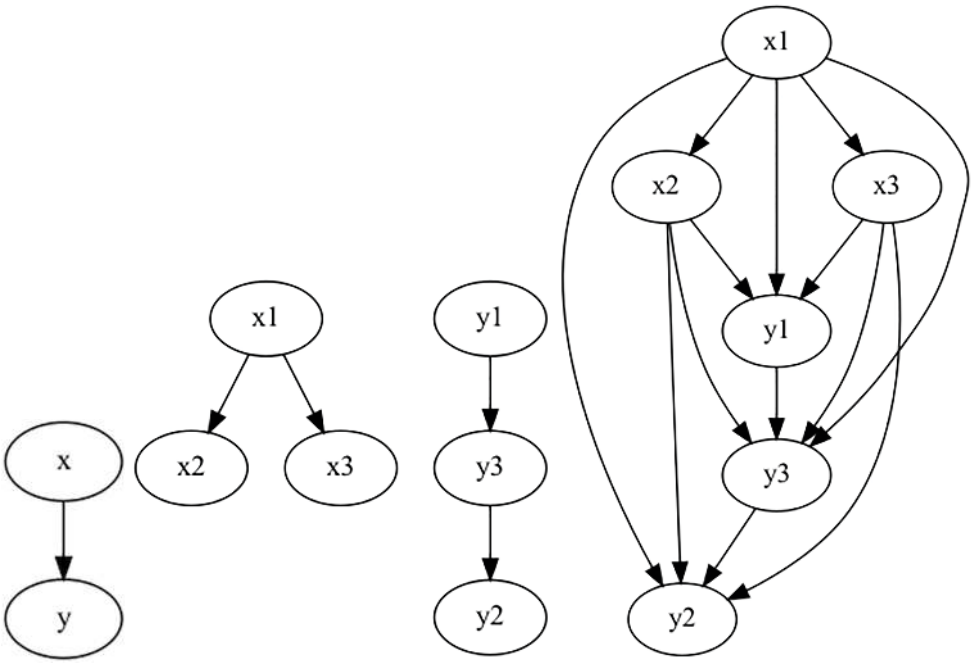

Fig. 14.1 A visualization of the last step combining all the groups into one unified graph. In this example, there are categories of features x and y. Each group is composed of three features. The DAG on the left models the inter-group causal relationships, the DAG in the middle models the intra-group causal relationships, and the DAG on the right is the final graph combining both results.

In this step, the inter-group causal relationships are fully connected, that is, if it is determined that $g_j \rightarrow g_k$ for some $g_j, g_k \in G_f$, it is not initially known which nodes (features) in g_j drive which nodes (features) in g_k. Thus, in order to not miss out on valid relationships, the authors elect to over-connect groups and filter out the irrelevant relationships in a step described in the Causal Validation section. Of course, an alternative approach here is to once again use an LLM to validate these emergent nodes, but the authors elect to go with an empirical causal validation approach for performance reasons.

In all the prompts, the authors include templates of the expected output format and describe the structure of the list of features in the input as these are best practices used to improve the quality of responses of GPT-4 [21]. The authors will omit mentioning this step for the rest of the paper.

Now, here is a detailed description of the necessary steps to generate a valid causal DAG for each group of features. These steps are also used to generate the final causal DAG used to link each group together; the sample prompts for this process can be seen in Appendix B:

(1) Generate the set of all possible valid causal relationships for a group of features using GPT-4 and model this as an undirected graph.
(2) Generate a causal DAG from the set of causal relationships obtained in the previous step by determining the directions of all the causal relationships using GPT-4.
(3) Verify the validity of all the causal relationships and correct any mistakes made during the initial generation of the DAG using GPT-4.
(4) Verify the corrections made were valid and verify if there are any remaining mistakes in the DAG using GPT-4.
(5) If there were any final mistakes, correct these mistakes and return the final version of the causal DAG.

The authors describe this process in three distinct phases: Causal Exploration, Causal Inference, and Causal Validation.

14.4.1. *Causal exploration*

During causal exploration, the causal DAGs are repeatedly generated over the same group of features in order to generate the set of all possible causal relationships for the given group. The DAGs were generated 10 times per group for the experiments described in this paper. This step is necessary to reduce the impact of the non-deterministic behavior of LLMs.

For each iteration of causal DAG generation, the authors provide the following definition of a causal relationship: *A causal relationship between two features means that a change in one feature causes a change in the other feature*. The authors also use general chain-of-thought instructions to instruct GPT-4 [20] to include the feature's definitions and the definition of causal relationship in its reasoning. The authors use in-context learning and justifications by example and thought processes as part of the prompt engineering process to generate high-quality outputs.

Sample prompts for these steps can be seen in Appendix Section C.1.

14.4.2. *Causal inference*

During causal inference, the authors include the set of edges in the prompt used to form the initial undirected graph generated during the causal exploration phase. These are outputted as a string that can be parsed into a list of tuples. Finally, the authors use the same techniques as in the previous phase and also include the definitions of a DAG and a causal relationship in the prompts.

Now for the set of edges, the authors prompt the LLM to generate a causal DAG. GPT-4 performs perhaps surprisingly well for this prompt and the prompt above, which in combination return a DAG based on definitions of the input features F forming the nodes in the graph. The LLM is also prompted to return a justification for each decision it makes when constructing the graph. Thus, the LLM returns a justification for the direction of each edge; the authors observe that including the justification also generates higher-quality outputs, similar to the approach seen in Wei et al. [20].

Sample prompts for these steps can be seen in Appendix Section C.2.

14.4.3. *Causal validation*

During the causal validation phase, the goal is to correct any mistakes the LLM makes have made during its initial DAG generation. The authors therefore use all the same techniques mentioned in the previous two phases in order to improve performance.

The authors also include the outputs of previous prompts in order to provide context and generated justification connected to the previous outputs. This history starts at the causal inference phase. Thus, before performing validation, the prompt includes the initial undirected graph representation, the features names and definitions, the initial DAG generation, and the justifications for each edges direction in the DAG. An example of this final validation prompt can be seen in Appendix Section C.3.

Now there are three main types of mistakes an LLM can make during graph generation:

(1) False negative/missing causal relationship.
(2) False positive causal relationship (i.e., produce a nonexistent/ illogical relationship).
(3) Produce a correct causal relationship with an incorrect causal direction.

The authors include this in the verification prompt and prompt the LLM to analyze the whole causal DAG generation process and attempt to remedy any mistakes made in this process. If so, the LLM returns a list of the invalid edges and a list of the corrections, both accompanied with justifications for their inclusions. The DAG is then corrected using the parsed responses before starting the final verification step.

During this final verification step, the authors also add to the concatenation of previous correction prompts (i.e., the "chat history") if any corrections needed to be made. The LLM is prompted to "verify its thought process" iteratively (in a loop) and apply corrections where necessary until the stopping criteria pre-specified within the prompt itself is hit.

Given the validity of the causal DAG, the authors use the do-calculus from observation data to further validate the causal relationship non-parametrically. Do-calculus provides a statistical significant value for each edge. Edges that are not statistically significant are removed from the DAG. In addition, a cross-validation is performed on a subset of the observation data to confirm the statistical significance of the edge. This final step is called the refutation step. The following definition captures the requirement that a causal query Q within the factors be estimable from the factor exposure:

Definition 14.4.1 *A causal query Q(M) is identifiable from a graph $\mathscr{G}$, given a set of assumptions A and the set of observed factors V, if for any pair of models M_1 and M_2 that satisfy A, we have*

$$P_{M_1}(v) = P_{M_2}(v) \Rightarrow Q(M_1) = Q(M_2).$$

When a query Q is given in the form of a *do*-expression, for instance, $Q = P(y|do(x), z)$, its identifiability can be decided systematically using an axiomatic system known as the *do*-calculus [22]. The axiomatic system replaces probability formulas containing the *do*-operator with ordinary conditional probabilities in three axiom schemes.

Let X, Y, Z, and W be arbitrary disjoint sets of nodes in a casual DAG $\mathscr{G}$. Denoting $\mathscr{G}_{\overline{\mathscr{X}}}$ the graph obtained from $\mathscr{G}$ by deleting all arrows pointing to nodes in X, $\mathscr{G}_{\underline{\mathscr{X}}}$ the graph obtained from $\mathscr{G}$ by removing all arrows emerging from nodes in X. The following three rules are valid for every interventional distribution compatible with $\mathscr{G}$.

Rule 1 (Insertion/deletion of observations):

$$P(y|do(x), z, w) = P(y|do(x), w) \text{ if } (Y \perp\!\!\!\perp Z|X, W)_{\mathscr{G}_{\overline{X}}}$$

Rule 2 (Action/observation exchange):

$$P(y|do(x), do(z), w) = P(y|do(x), z, w) \text{ if } (Y \perp\!\!\!\perp Z|X, W)_{\mathscr{G}_{\overline{X}\underline{Z}}}$$

Rule 3 (Insertion/deletion of actions):

$$P(y|do(x), do(z), w) = P(y|do(x), w) \text{ if } (Y \perp\!\!\!\perp Z|X, W)_{\overline{X}\overline{Z(W)}}$$

where $Z(W)$ is the set of Z-nodes that are not ancestors of any W-node in $\mathscr{G}_{\overline{X}}$.

Note that in cases where the set X is empty, Rule 1 corresponds to the application of d-separation principles to interventional distributions, and Rule 2 becomes analogous to the backdoor adjustment criterion.

To establish identifiability of a query Q of factors, the rules of *do*-calculus are repeatedly applied to Q, until the final expression no longer contains a *do*-operator. The final *do*-free expression can serve as an estimator of Q. The do calculus was proven to be complete to the identifiability of causal effect [23, 24]. As such, the causality relationship generated by LLM is endorsed by observation data.

The authors also utilize L_1 regularization (Lasso Regression) when fitting the final models as a final causal validation/feature selection step, reducing the number of features used in the final model and reducing the multicollinearity of the final feature set. Since the Lasso model is fit on each training fold in a backtest, the final feature selection/causal validation step is dynamic for each period.

14.4.4. *LLMs for feature engineering*

In order to represent the relationships between the different features (edges) in the DAG as actual features, LLMs can also be used to select the most logical transformation of the two features composing each edge to create a new feature.

A simple approach for this is to provide a set of "primitives" or standard transformations and query an LLM to select one of the definitions of the features and the direction of the causal relationship between the two features.

This step could possibly be improved by using deep learning/LLM-based feature engineering frameworks or the newly proposed CAAFE framework cited in the literature review, as seen in Hollmann et al. [16].

14.5. Applications

The authors focus on two key applications of the framework outlined above: causal factor investing and the application of the DAG generation framework to other feature sets (using credit spread modeling as an example). In the following section, all the do-calculus computations on the DAG are done through the use of the following Python package: DoWhy [25]. The authors omit mentioning its use through the rest of the results section. The details of each are outlined below.

14.5.1. *Applications to causal factor analysis*

Factor investing is a financial theory aiming to explain the returns of stock as connected to risk *factors* that play a key role in risk management (where factor loads in a portfolio are carefully managed) and in investing strategies that aim to strategically load up on specific factors (e.g., "smart beta"). A researcher usually determines this set of factors by making causal assumptions and calculating factor exposures concerning the stock's returns using procedures inspired by Fama and French [3] and MacBeth [26] as mentioned in Prado [2].

Now even if these causal assumptions drive the subsequent modeling decisions the researcher will make in his analysis, they are rarely explicitly stated. Clearly stating these causal assumptions would require the researchers to provide empirical evidence to back their claims. Nevertheless, this opens up the opportunity to use the rules of do-calculus to de-bias their data and improve the explainability of the modeling structure, especially when using a large set of factors. Consequently, the proposed framework bridges this gap by automating this potentially incredibly laborious process.

The authors investigate the set of 153 factors presented in Jensen et al. [27]. Using the base definitions and names presented for each factor, the authors manually improve the definitions in order to make them more verbose and representative of their construction (another step that could potentially be automated/improved using LLMs). The time period analyzed is from November 30, 1971, to December 30, 2022. Each factor has an associated ETF modeling its daily returns over this period.

Now the authors compare the causal DAGs generated using their proposed LLMs clustering method to a classical correlation clustering algorithm (described in Section 14.3.2). They generate a first clustering using an LLM (GPT-4) and compare it to a baseline correlation clustering approach, as seen in, for example, Jensen et al. [27] using hierarchical agglomerative clustering. The authors in Jensen et al. [27] define the distance between factors in the second clustering as one minus their pairwise correlation and use Ward as the linkage criterion. The authors also define the correlation based on CAPM-residual returns of US factors signed as in the original paper by Hou et al. [28]. Figure 14.2 compares inter-cluster correlations of the two methods; although our proposed method naturally has lower inter-cluster correlations, the causal DAGs for each cluster are substantially more cohesive from an expert judgment perspective, as can be seen in this GitHub repository. Despite not using correlation or time-series data as an input, the LLM clusters produce average inter-cluster correlations of 0.292, compared to 0.543 for the correlation clustering approach, and intra-cluster correlations of 0.0527 compared to 0.0047. The clusters produced by the LLMs are also in some ways more orthogonal, having average absolute inter-cluster correlations of 0.086, compared to 0.1451 for the correlation clustering approach. The authors then generate a complete DAG for each set of clusters ("causal" LLM clusters or correlation clusters) as described in Section 14.3.

Now in order to confirm the causal DAGs using do-calculus, the ETF returns are used to construct proxies for the factors of each individual stock in the S&P 500 by computing the rolling beta exposures of each stock to the 153 factor ETFs. This generates a feature set X_t of 500 stocks and 153 features per stock per period, with each feature being a proxy for a factor exposure computed using an individual ETF. The authors then use these proxies for factor exposure to verify all the causal relationships in each causal DAGs generated by the LLM edge-wise. In other words, they use do-calculus to confirm the validity of edges in the causal DAGs. However, due to computational constraints, the authors did not run the do-calculus on the fully connected graph as in Figure 14.1. Indeed, they ran do-calculus separately on each DAG modeling the intra-group causal relationships.

In the case of the causal DAGs modeling the inter-group causal relationships, the authors executed the do-calculus using the first n principal components sufficient enough to explain at least 85% of the groups variance. If any of the components is proven invalid by the do-calculus computation, all associated edges are removed. Now, having validated all the causal DAGs, the authors add a new node, each of them representing the stock returns. An example visualization can be seen in Fig. 14.3. For each causal DAG representing the intra-group causal relationships, the authors then use do-calculus to analyze the causal relationships of each factor with respect to the S&P 500 constituents stock returns. The authors remove from the causal DAGs any causal relationship/edge to the stock returns that are not statistically significant with a p-value of 0.05. These statistically insignificant relationships represent factors that do not directly impact S&P 500 constituents stock returns. Nevertheless, their values may still impact the stock returns through their causal relationships with other features in the DAGs.

For an average period, there are a total of 103.4 factors left out of the initial 153 factors, combined with 184.6 features representing the edges in the DAG (from a total of 192 across all of the LLM-cluster DAGs).

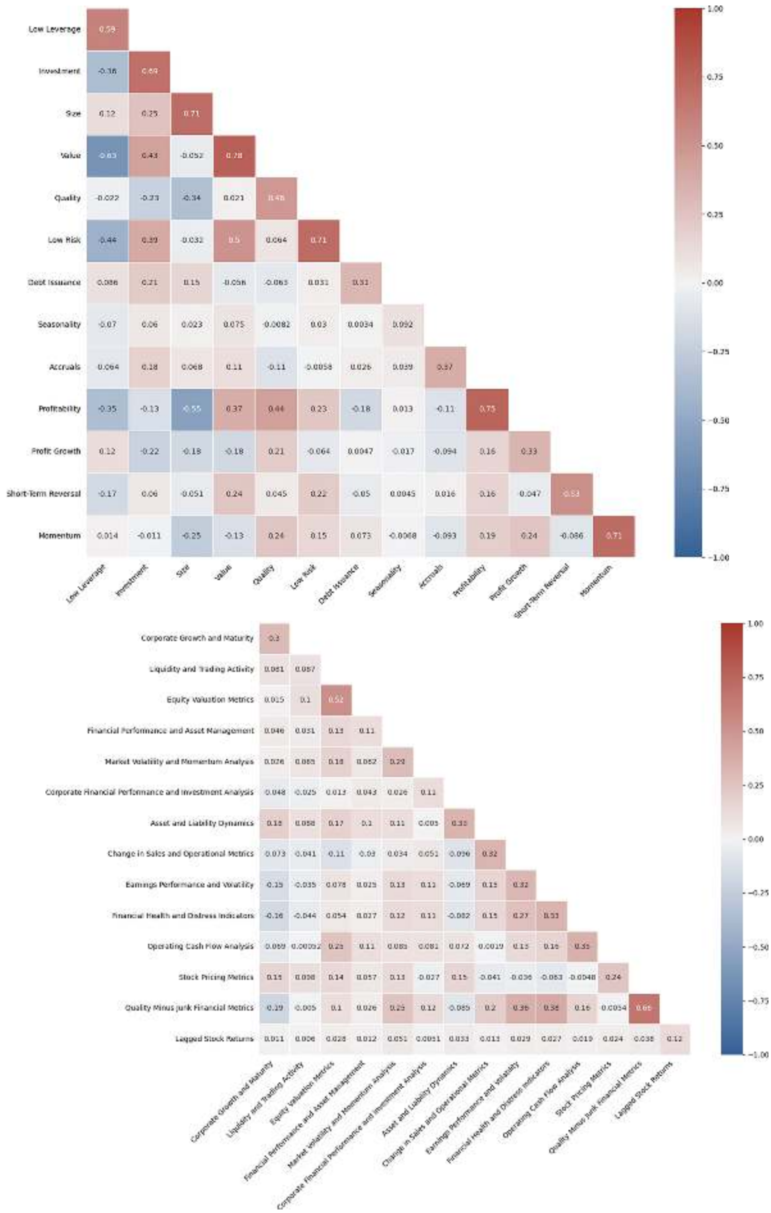

Fig. 14.2 This figure shows the average pairwise Pearson correlation between factors from different clusters and factors from the same clusters for hierarchical agglomerative clustering (left) and LLM clustering (right) using data on US stocks from 1971 to 2021. The figure on the left is inspired from Jensen et al. [27].

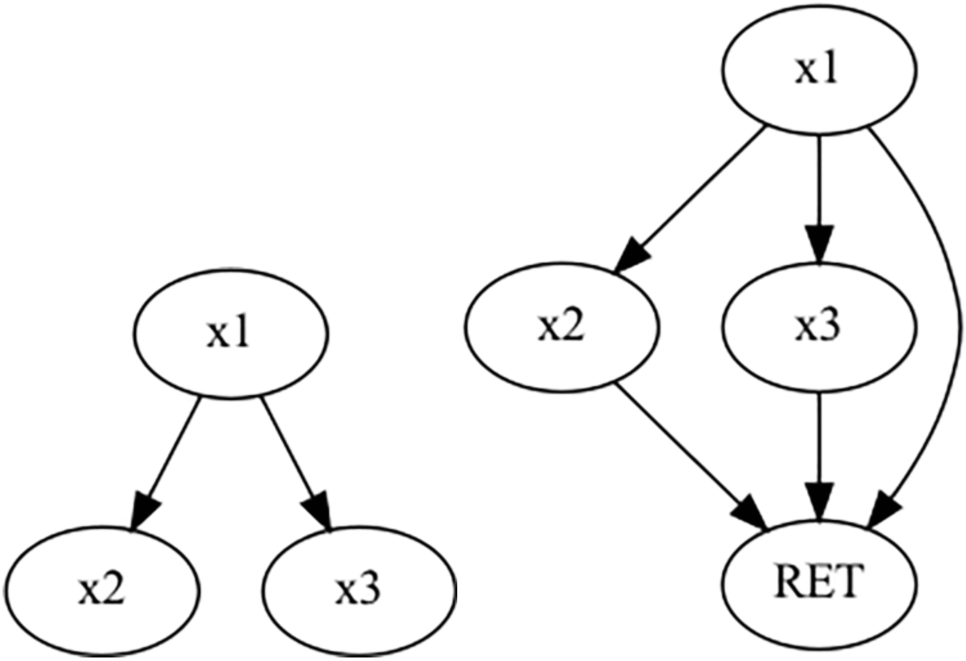

Fig. 14.3 This figure shows how the authors connect each feature to the Y node of interest. In the factor investing case, this Y node is represented by the stock returns RET.

A high-level comparison of the two clustering approaches can be Seen in Table 14.1. In addition, a more detailed analysis comparing the average performance per each of the hierarchical clusters as seen in Jensen et al. [27] is provided in Table 14.2, and of the LLM-clusters in Table 14.3. One can see that the LLM clusters provide a comparable level of predictive power for monthly returns, while providing a much less correlated and more interpretable feature set. The LLM clusters also utilize more edges per cluster, leading to a potentially richer representation of causal relationships. In addition, the percentage of the refuted nodes are much lower than in the combined dataset, given there is less redundancy within the individual clusters.

Table 14.1 A comparison of the DAGs generated using hierarchical and LLM-based clustering.

Clustering methodology	Average number of nodes	Average number of edges	Refuted nodes (%)	Refuted edges (%)
Correlation Clusters	10.93	13.71	3.64	31.29
LLM Clusters	11.77	11.54	3.15	32.43

Table 14.2 Summary statistics of in- and out-of-sample RMSE for each of the correlation clusters inspired by Jensen et al. [27].

	Avg RMSE (out-of-sample)	Avg R^2 (in sample)	Avg adjusted R^2 (in sample)	Avg number of nodes used	Total number of nodes on graph	Avg number of edges used	Total number of edges
Low leverage	0.0997	0.0496	0.0465	10.6818	11	13.9766	14
Debt issuance	0.0971	0.0290	0.0269	6.6703	7	10.9318	11
Profitability	0.1001	0.0468	0.0436	10.6993	11	14.9923	15
Short-term reversal	0.0972	0.0260	0.0247	5.6987	6	4.9538	5
Low risk	0.1006	0.0683	0.0641	17.6613	18	15.9772	16
Investment	0.1027	0.0645	0.0601	21.6206	22	17	17
Size	0.0958	0.0251	0.0241	4.7000	5	3.9856	4
Accruals	0.0967	0.0251	0.0235	5.7115	6	6.9643	7
Value	0.1012	0.0579	0.0540	17.6442	18	14.9987	15
Quality	0.0987	0.0571	0.0540	16.6799	17	10	10
Momentum	0.1044	0.0509	0.0474	7.6683	8	18.9626	19
Profit growth	0.0978	0.0419	0.0394	11.6651	12	9.9462	10
Seasonality	0.0973	0.0467	0.0446	11.6827	12	6.9011	7
Average	0.0992	0.0453	0.0425	11.4449	11.7692	11.5069	11.5385

Table 14.3 Summary statistics of in- and out-of-sample RMSE for each of the LLM clusters.

	Avg RMSE (out-of-sample)	**Avg R^2 (in sample)**	**Avg adjusted R^2 (in sample)**	**Avg number of nodes used**	**Total number of nodes on graph**	**Avg number of edges used**	**Total number of edges**
Corporate growth and maturity	0.0950	0.0122	0.0119	1.6827	2	0.9615	1
Quality minus junk financial metrics	0.0957	0.0205	0.0196	3.7452	4	3	3
Stock pricing metrics	0.0949	0.0121	0.0118	1.6731	2	1	1
Asset and liability dynamics	0.0999	0.0587	0.0546	18.6346	19	17.9647	18
Financial health and distress indicators	0.0977	0.0347	0.0329	4.7423	5	8.9744	9
Lagged stock returns	0.0993	0.0521	0.0490	9.7135	10	15.8894	16
Financial performance and asset management	0.1005	0.0599	0.0558	17.6528	18	16.9717	17
Change in sales and operational metrics	0.0957	0.0173	0.0161	3.6779	4	5.9135	6
Corporate financial performance and investment analysis	0.1010	0.0597	0.0549	15.6611	16	22.9791	23
Market volatility and momentum analysis	0.1181	0.0992	0.0900	26.6546	27	48.9714	49
Operating cash flow analysis	0.0965	0.0279	0.0265	5.6955	6	5.9744	6
Liquidity and trading activity	0.0999	0.0534	0.0507	11.6731	12	10.9773	11
Earnings performance and volatility	0.1002	0.0509	0.0473	11.6907	12	16.9604	17
Equity valuation metrics	0.1005	0.0547	0.0509	15.7212	16	14.9526	15
Average	0.0996	0.0438	0.0409	10.6156	10.9286	13.6779	13.7143

14.5.2. *Applications to causal modeling of credit spread drivers*

The drivers of changes in credit spread are a long-lasting argument. Early studies by Black and Scholes [29] and Merton [30] introduced a structural model to explain corporate default risk and inspired the search for company-level and macro variables; later, Collin-Dufresn et al. [31] stated the impact of market-related factors, and Chen et al. [32] identified the high correlation between the equity market performance and the spread. But is there any causality relationship between these discovered factors? If so, the redundant features can be removed using a causality approach.

To identify the causal relationship within the factors that might drive the credit spread, the authors utilize the dataset Gilchrist and Zakrajšek [33] and generate the DAG following the steps mentioned in the methodology. This numerical dataset consists of different macroeconomic factors and market-related factors. The data consists of monthly and quarterly data. Each edge in the DAG is validated using the rules of do-calculus. Table A14.1 presents the statistical significance of the rejection of causal relationships within the edges of the DAG.

In Table A14.1, each edge is associated with its effect (i.e., strength of the relationship), its p-value and its refutation p-value. The p-value represents the statistical significance of the edge and the refutation p-value represents the robustness of the p-value associated with the effect. Indeed, the refutation p-value is computed using K-Fold Stratification of the data to test the degree of randomness associated with the edge. The authors reject/remove any edge who's p-value is greater than 0.05 as shown in Figure A14.2. Any edge where the p-value is significant (i.e., smaller than 0.05), but whose refutation p-value is significant, is rejected as well. The authors will refer to the steps described as validating the causal DAG *edgewise*.

The authors reject 39 edges using do-calculus out of the 89 edges that GPT-4 returned. This is within the expectation as the LLM DAG generation approach focused mainly on recall and is expected to generate denser initial graphs. In addition, the dataset is much smaller than the dataset used in the previous section, leading to more frequent rejections.

Using the framework, the authors are able to identify the causality relationship within the drivers of credit spread; to a practitioner, this may be a good starting point for a feature set or a causal model, especially in situations where expert judgment in a given domain is not readily available in-house.

14.6. Limitations and Future Work

The authors leverage GPT-4 for all the analyses presented in this paper. In its current state, GPT-4 is a powerful language model that is able to produce causal relationships intuitive to human experts and those that align well with empirical observations.

Nevertheless, it has some key limitations researchers and practitioners should be aware of. One is in making unpredictable mistakes, which can be very costly in the causal DAG generation and similar use causes. Indeed, the authors of Kıcıman et al. [12] repeatedly mention this in their experiments on pair-wise causal discovery and full graph discovery that LLMs produce infrequent but unpredictable errors, making the process of automating causal discovery unfeasible for the moment. Thus, it is essential to have human supervision to correct the LLMs in the case they produce a hallucination (i.e., incoherent output). In order to make this process less laborious, an LLM can also be used to recursively correct its own mistakes (as can be seen in Appendix C). The authors of Kıcıman et al. [12] found that GPT-4 was the only model capable of verifying the consistency of its outputs (self-consistency), though the capabilities of LLMs are evolving rapidly, and it is possible other models can display similar capabilities with additional prompt engineering.

The authors also hope that the DAGs generated in the paper, which can be accessed on GitHub here, will spur further research into optimal representations for risk and factor models. This paper shows that causal approaches can now be scaled to provide a working alternative to correlation-based approaches for generating feature representations in cases where the universe of possible features is large and the signal-to-noise ratios are too low to find the optimal feature representations empirically.

These causal DAGs, similar to those demonstrated in this paper, can be leveraged to improve model robustness and out-of-sample performance, similar to the process of injecting expert judgment priors into predictive models. The authors are particularly interested in how these alternative feature representations can be used in constructing risk and factor models and how such models may compare with state-of-the-art models used by practitioners today.

It is also necessary to remember that the performance of the LLMs drastically relies on the prompt engineering process and the current state-of-the-art in foundational models. The authors believe that the end-to-end approach can be improved substantially both through injecting additional expert knowledge into the prompts and through the evolution of foundational models.

The authors believe that combining state-of-the-art LLMs and techniques that can be used to push their performance, as described in this paper, in combination with correlation-based methods as in Sharma and Kiciman [25], can be a fruitful area of research and a significant step toward automating end-to-end causal discovery in finance and beyond.

APPENDIX

A. Credit Spread Figures and Tables

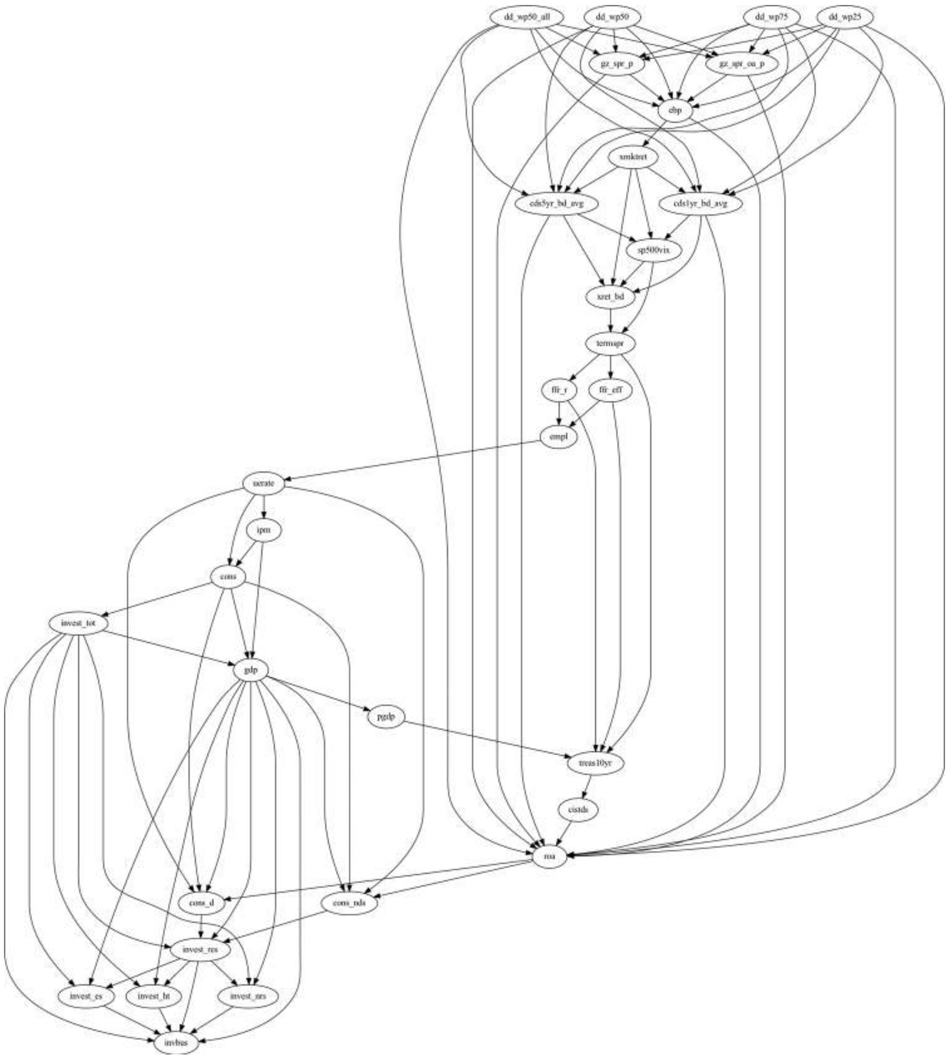

Fig. A14.1 This figure shows the credit spread DAG generated by the causal DAG generation framework before do-calculus.

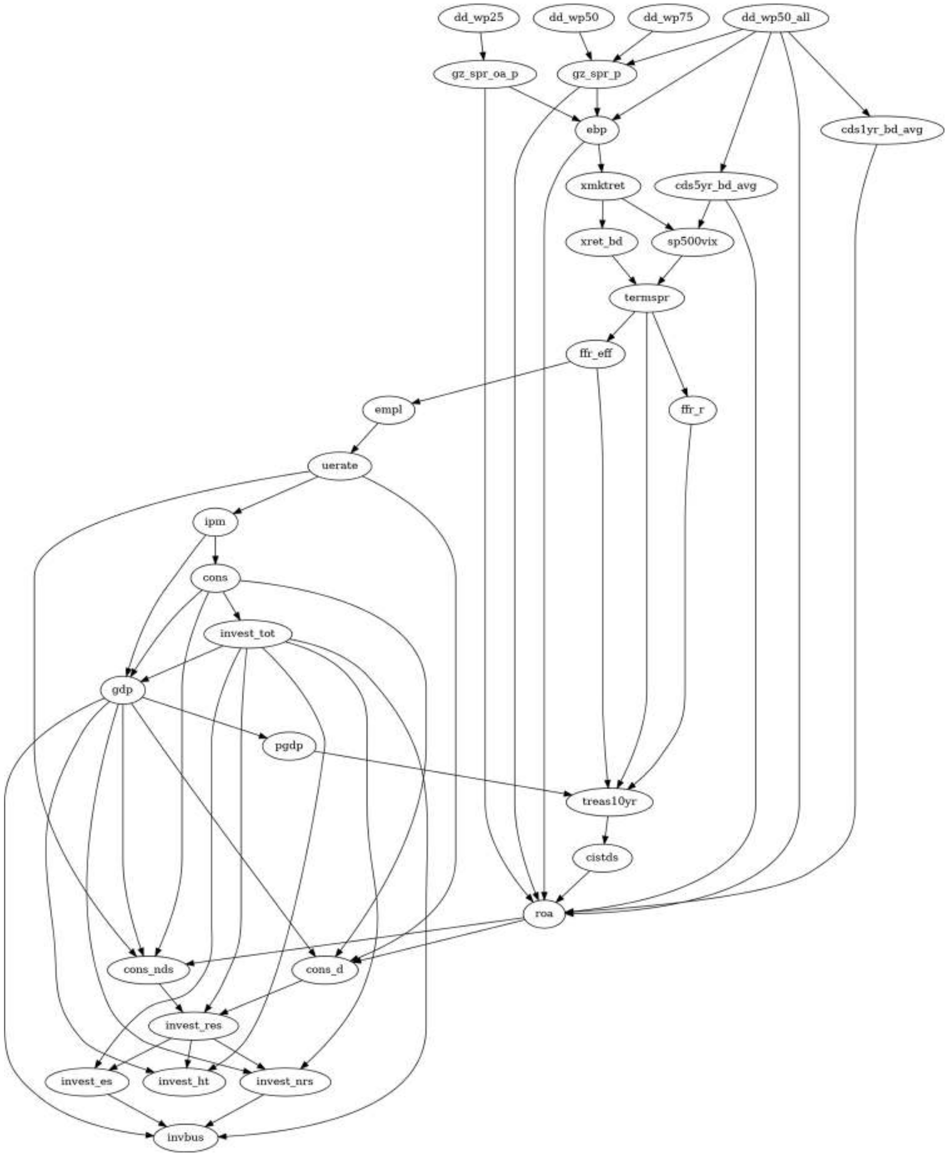

Fig. A14.2 The figure shows the process of how do-calculus removed the redundant edges within the LLM-generated causal DAG for credit spread. The original LLM-generated DAG can be found in Figure A14.1.

Table A14.1 Nonparametric causal estimation result for credit spread DAG. Each edge in the graph is designated by a start node and an end node. The effect consists of a measure of the strength of the causal relationship, the *p*-value signifies its level of significance, and the refutation *p*-value is a measure of how robust the *p*-value is by testing its significance through K-Fold stratification.

Start node	End node	Effect	*p*-value	Refutation *p*-value	Start node	End node	Effect	*p*-value	Refutation *p*-value
ebp	roa	–0.50	0.00	0.94	invest_res	invest_nrs	–0.15	0.00	0.84
ebp	xmktret	–2.20	0.00	1.00	dd_wp25	ebp	–0.29	0.16	0.96
xmktret	sp500vix	–0.36	0.00	1.00	dd_wp25	roa	0.19	0.55	0.70
xmktret	xret_bd	1.46	0.00	0.98	dd_wp25	gz_spr_p	–0.42	0.05	0.54
xmktret	cds1yr_bd_avg	0.02	0.05	0.84	dd_wp25	gz_spr_oa_p	–0.01	0.00	0.78
xmktret	cds5yr_bd_avg	0.01	0.19	0.94	dd_wp25	cds1yr_bd_avg	–0.18	0.64	0.72
sp500vix	xret_bd	–0.08	0.51	0.98	dd_wp25	cds5yr_bd_avg	–0.12	0.91	0.92
sp500vix	termspr	–0.09	0.00	0.96	dd_wp50	ebp	–0.26	0.13	0.98
xret_bd	termspr	–0.04	0.03	0.70	dd_wp50	roa	0.11	0.99	0.98
termspr	treas10yr	0.23	0.00	0.90	dd_wp50	gz_spr_p	–0.45	0.00	0.82
termspr	ffr_r	1.10	0.00	0.96	dd_wp50	gz_spr_oa_p	–0.06	0.62	0.94
termspr	ffr_eff	1.27	0.00	0.94	dd_wp50	cds1yr_bd_avg	–0.18	0.39	0.86
empl	uerate	0.00	0.00	0.96	dd_wp50	cds5yr_bd_avg	–0.09	0.93	0.94

uerate	ipm	−2.63	0.00	0.80	dd_wp75	ebp	−0.07	0.99	0.94
uerate	cons	40.35	0.05	1.00	dd_wp75	roa	0.00	0.51	0.96
uerate	cons_nds	233.01	0.00	0.96	dd_wp75	gz_spr_p	−0.29	0.01	0.78
uerate	cons_d	32.00	0.00	1.00	dd_wp75	gz_spr_oa_p	−0.08	0.09	0.90
ipm	gdp	135.04	0.00	0.92	dd_wp75	cds1yr_bd_avg	−0.06	0.93	0.80
ipm	cons	97.26	0.00	0.94	dd_wp75	cds5yr_bd_avg	−0.04	0.85	0.94
cons	gdp	1.25	0.00	0.86	dd_wp50_all	ebp	−0.47	0.00	0.86
cons	invest_tot	0.28	0.00	0.80	dd_wp50_all	roa	0.46	0.00	0.78
cons	cons_nds	0.87	0.00	0.92	dd_wp50_all	gz_spr_p	−0.54	0.00	0.56
cons	cons_d	0.11	0.00	0.90	dd_wp50_all	gz_spr_oa_p	−0.03	0.48	0.98
invest_tot	invest_res	1.02	0.00	0.86	dd_wp50_all	cds1yr_bd_avg	−0.41	0.00	0.90
invest_tot	gdp	0.63	0.00	0.90	dd_wp50_all	cds5yr_bd_avg	−0.35	0.00	0.98
invest_tot	invbus	0.44	0.00	0.88	gz_spr_p	roa	−0.84	0.00	0.98

(Continued)

Table A14.1 Nonparametric causal estimation result for credit spread DAG. Each edge in the graph is designated by a start node and an end node. The effect consists of a measure of the strength of the causal relationship, the *p*-value signifies its level of significance, and the refutation *p*-value is a measure of how robust the *p*-value is by testing its significance through K-Fold stratification. (*Continued*)

Start node	End node	Effect	*p*-value	Refutation *p*-value	Start node	End node	Effect	*p*-value	Refutation *p*-value
invest_tot	invest_es	0.76	0.00	0.80	gz_spr_p	ebp	0.08	0.00	0.74
invest_tot	invest_ht	–0.26	0.00	0.88	gz_spr_oa_p	roa	1.69	0.00	0.88
invest_tot	invest_nrs	0.39	0.00	0.88	gz_spr_oa_p	ebp	–2.46	0.00	0.88
gdp	invbus	–0.20	0.00	1.00	cds1yr_bd_avg	sp500vix	–4.53	0.22	1.00
gdp	invest_res	0.03	0.81	0.80	cds1yr_bd_avg	xret_bd	–1.23	0.51	0.98
gdp	pgdp	0.01	0.00	0.98	cds1yr_bd_avg	roa	–1.27	0.00	0.94
gdp	cons_nds	–0.08	0.03	0.98	cds5yr_bd_avg	sp500vix	16.35	0.00	0.94
gdp	cons_d	0.09	0.02	0.80	cds5yr_bd_avg	xret_bd	–1.77	0.37	0.94
gdp	invest_es	0.04	0.24	0.94	cds5yr_bd_avg	roa	–1.43	0.00	0.68
gdp	invest_ht	0.22	0.00	0.96	ffr_r	empl	691.27	0.06	0.98
gdp	invest_nrs	–0.21	0.00	0.76	ffr_r	treas10yr	0.44	0.00	0.96
pgdp	treas10yr	–0.14	0.00	1.00	ffr_eff	empl	3683.78	0.00	0.76
treas10yr	cistds	–21.59	0.00	0.94	ffr_eff	treas10yr	0.36	0.00	0.86
cistds	roa	–0.02	0.00	0.76	cons_nds	invest_res	–0.82	0.00	0.86
roa	cons_nds	36.22	0.00	0.78	cons_d	invest_res	2.11	0.00	0.96
roa	cons_d	35.81	0.00	0.86	invest_es	invbus	–0.76	0.00	0.90
invest_res	invbus	0.02	0.27	0.98	invest_ht	invbus	–2.60	0.47	0.98
invest_res	invest_es	0.31	0.00	0.98	invest_nrs	invbus	–0.96	0.00	0.96

B. LLM Clustering Prompts

Fig. B14.1 Example of the first part of prompt 1 in the LLM clustering method. This prompts generates the initial clustering of features into mutually exclusive groups G_1.

Fig. B14.2 This prompt generated names and descriptions for groups in G_1 for subsequent regrouping.

Fig. B14.3 This prompt sub-clusters each group $g \in G_1$ into a sub-grouping S_g.

Fig. B14.4 The authors then generate names and description of the sub-groups S_g to allow for re-clustering of all sub-groups.

Fig. B14.5 This prompt generates the near-final grouping G_f by re-clustering all the sub-groups s, for each group g, into a brand new high-level grouping (now generated from sub-groups s rather than input features in F).

Fig. B14.6 Groups in G_f are also given names and detailed descriptions to enable final validation.

Fig. B14.7 The final grouping G_f is generated with a prompt aimed at identifying and remedying any "mistakes" in the final clustering by performing some additional merging of clusters in G_f.

C. DAG Generation Prompts

C.1. *Causal exploration prompts*

C.2. *Causal inference and causal validation prompts*

Fig. C14.1 This prompt produces candidate edges for each DAG G, by initially producing an undirected graph. It is applied once per cluster generated in the previous section.

Fig. C14.2 This prompt refines the candidate edges, adding directionality to the relationships.

Fig. C14.3 This prompt is applied recursively to identify any logical errors in the initial DAG, running until the stopping criteria of the LLM not suggesting any more changes is reached.

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